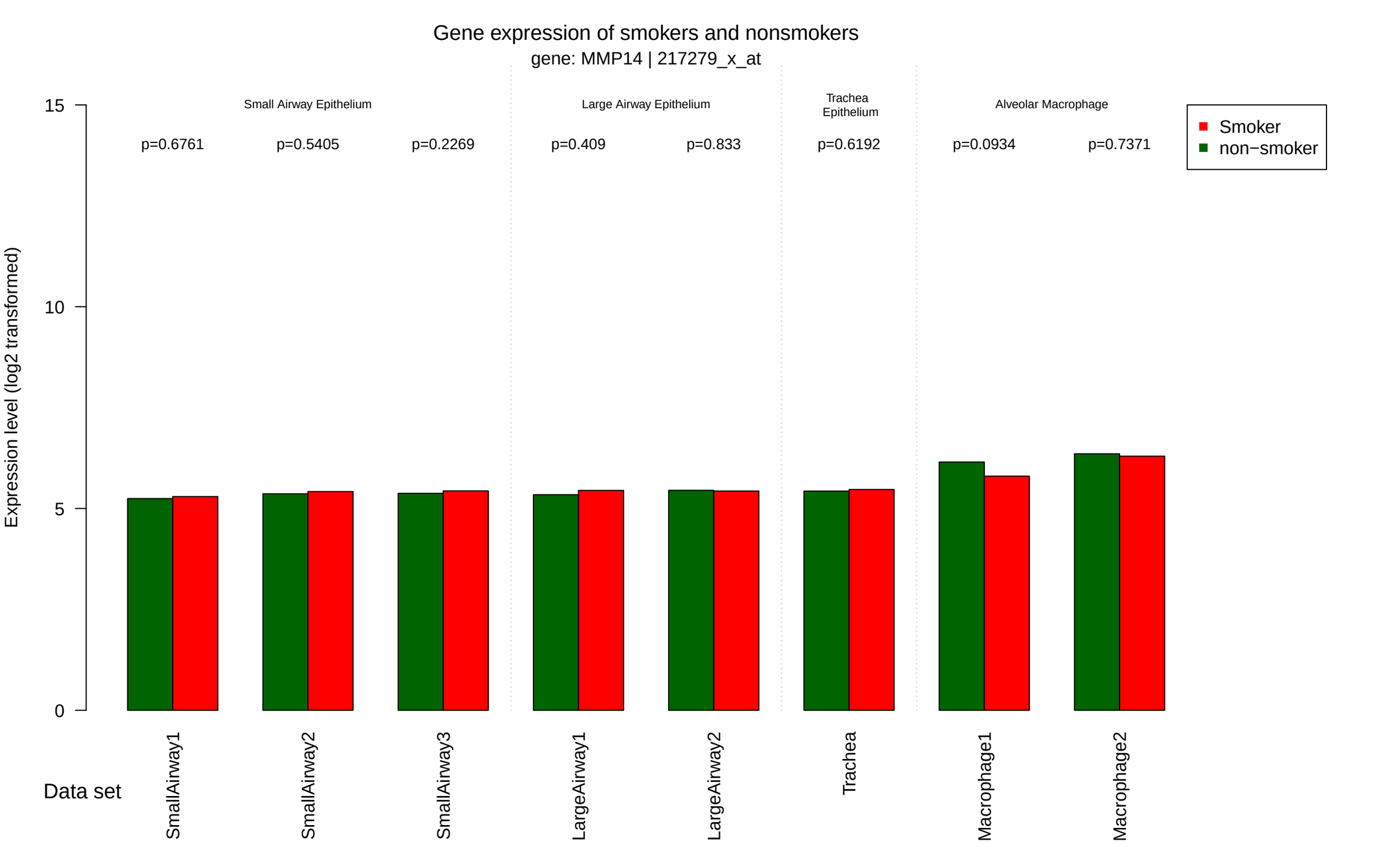
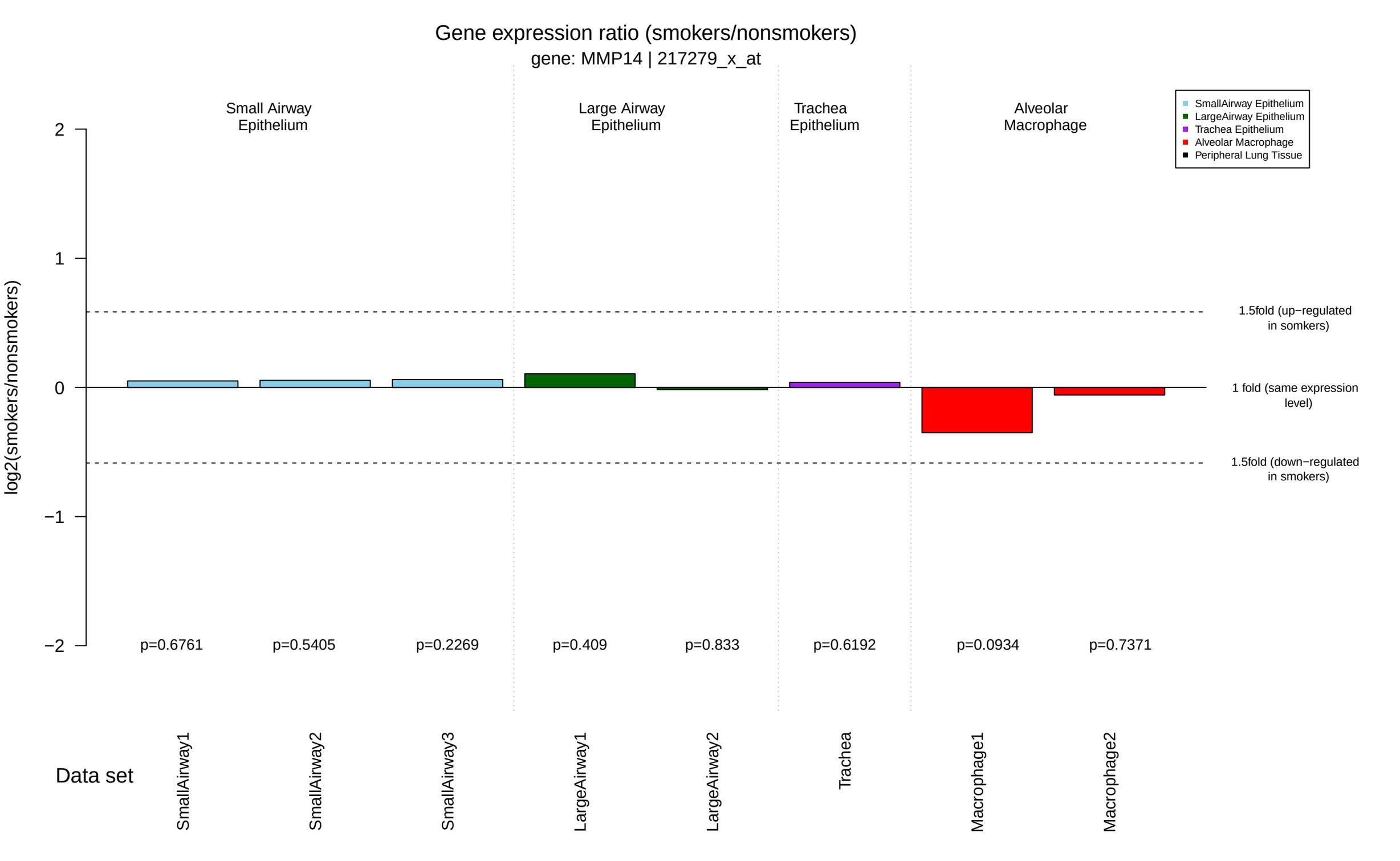
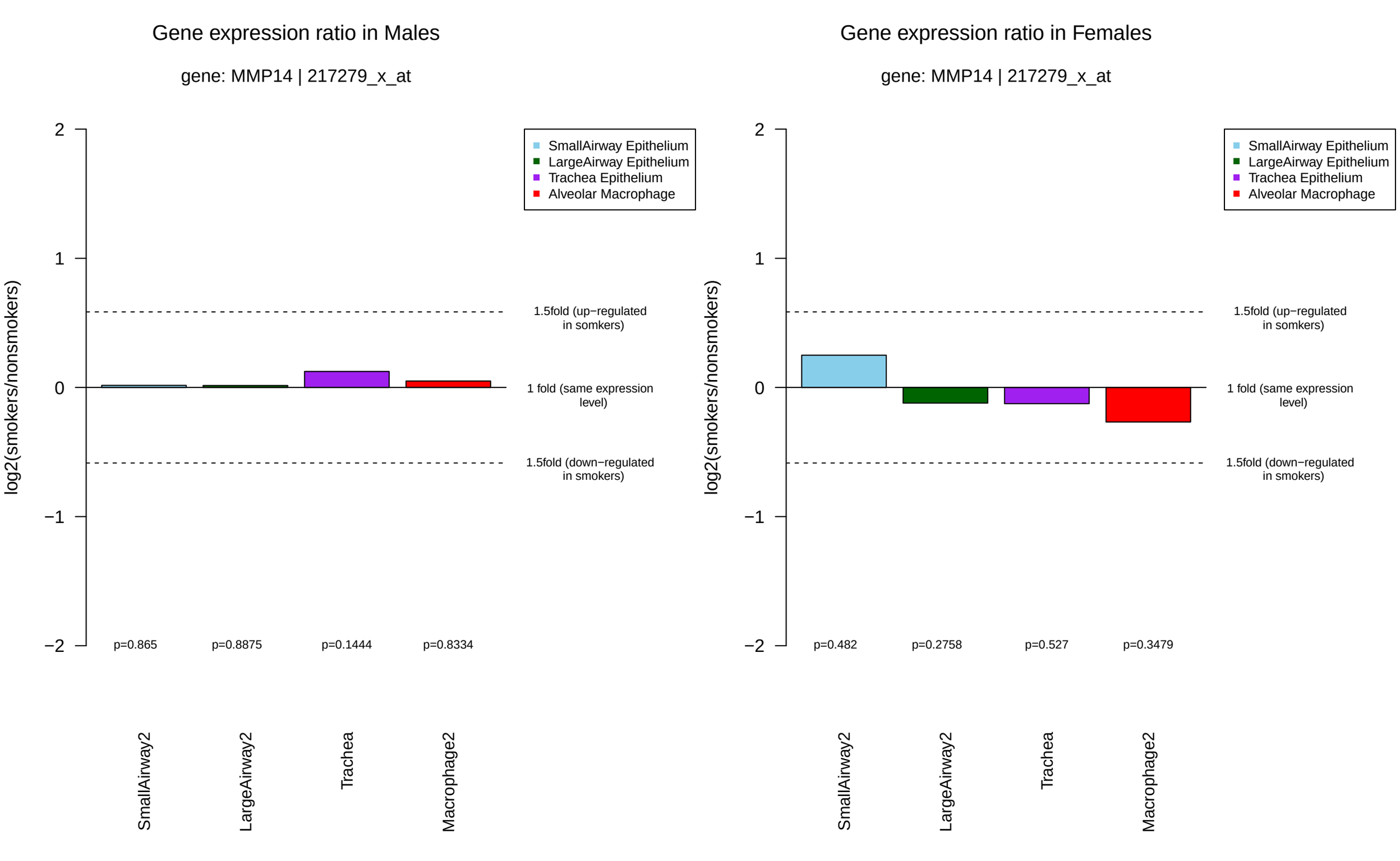
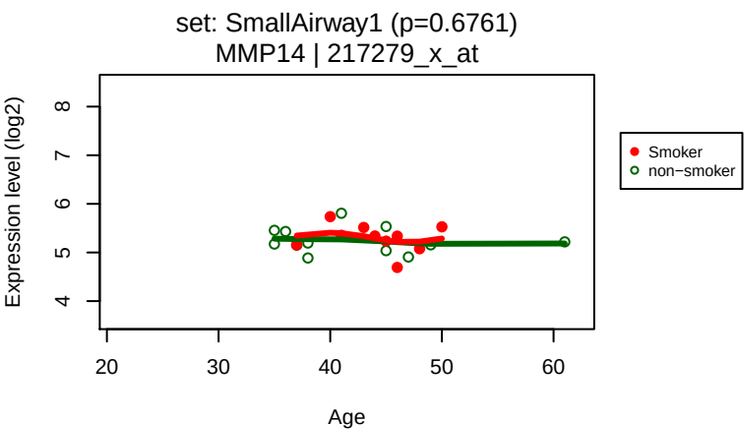
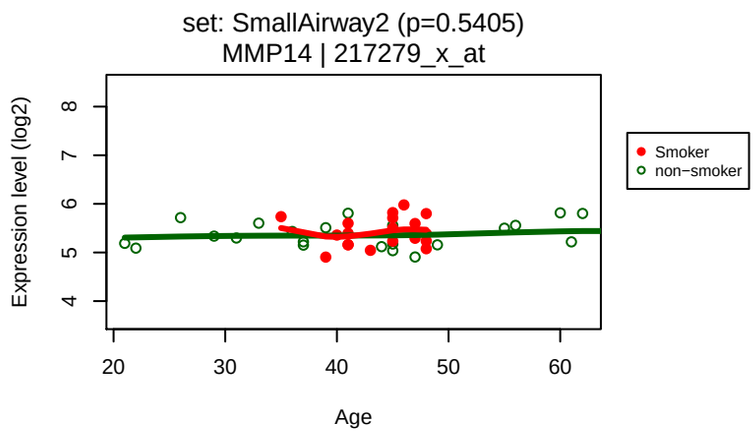
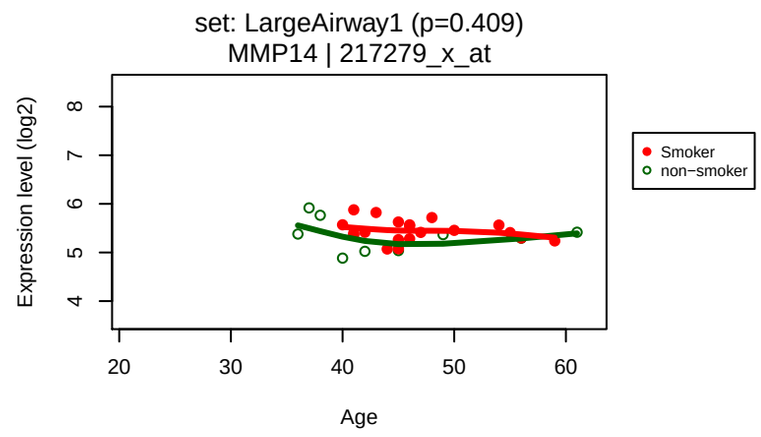

Gene expression of smokers and nonsmokers
gene: MMP14 | 217279_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6761
p=0.5405
p=0.2269
p=0.409
p=0.833
p=0.6192
p=0.0934
p=0.7371
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

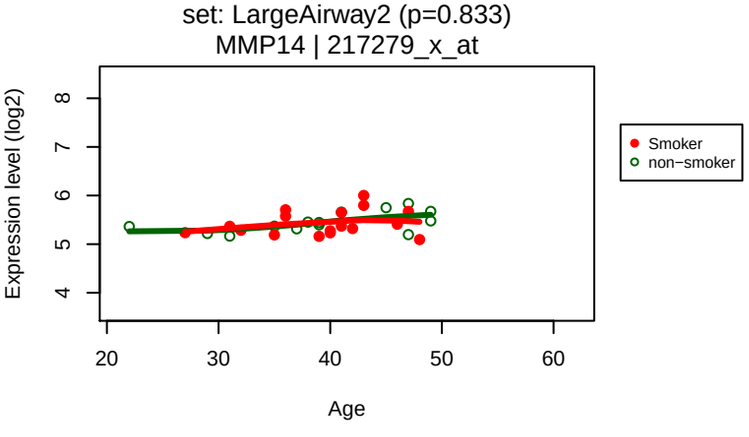

Gene expression ratio (smokers/nonsmokers)
gene: MMP14 | 217279_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6761
p=0.5405
p=0.2269
p=0.409
p=0.833
p=0.6192
p=0.0934
p=0.7371
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

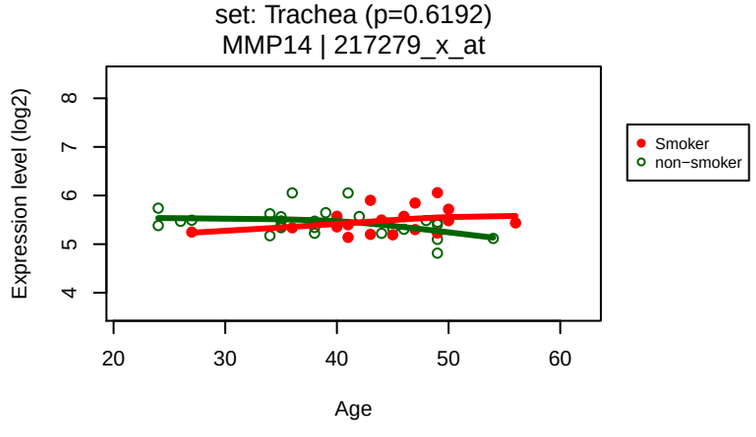

Gene expression ratio in Males
gene: MMP14 | 217279_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.865
p=0.8875
p=0.1444
p=0.8334
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MMP14 | 217279_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.482
p=0.2758
p=0.527
p=0.3479
SmallAirway2
LargeAirway2
Trachea
Macrophage2

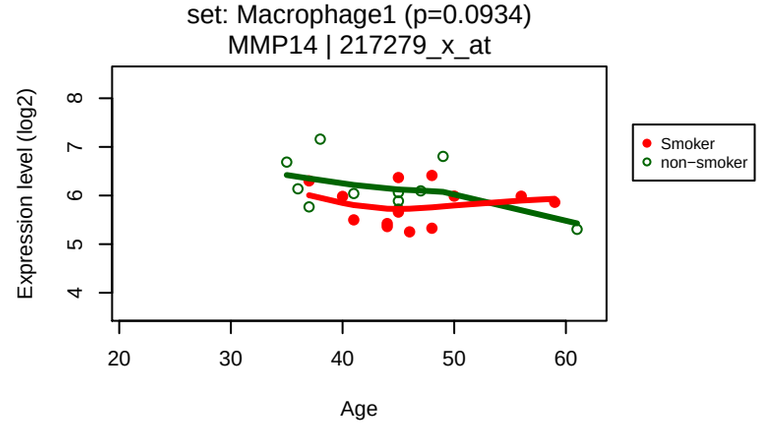

set: SmallAirway1 (p=0.6761)
MMP14 | 217279_x_at
Expression level (log2)
Age
Smoker
non−smoker

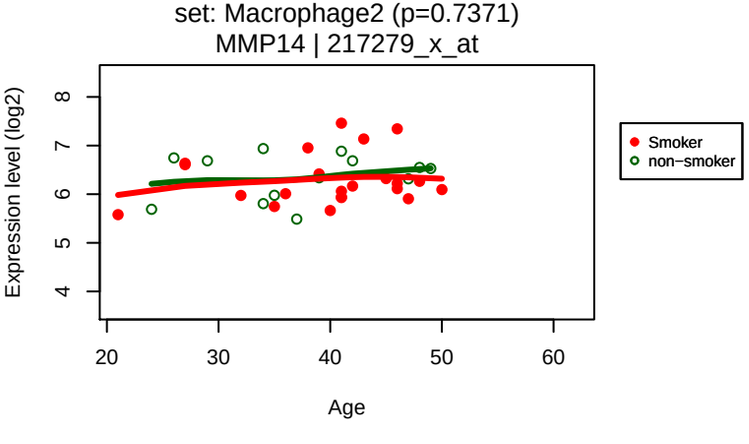

set: SmallAirway2 (p=0.5405)
MMP14 | 217279_x_at
Expression level (log2)
Age
Smoker
non−smoker

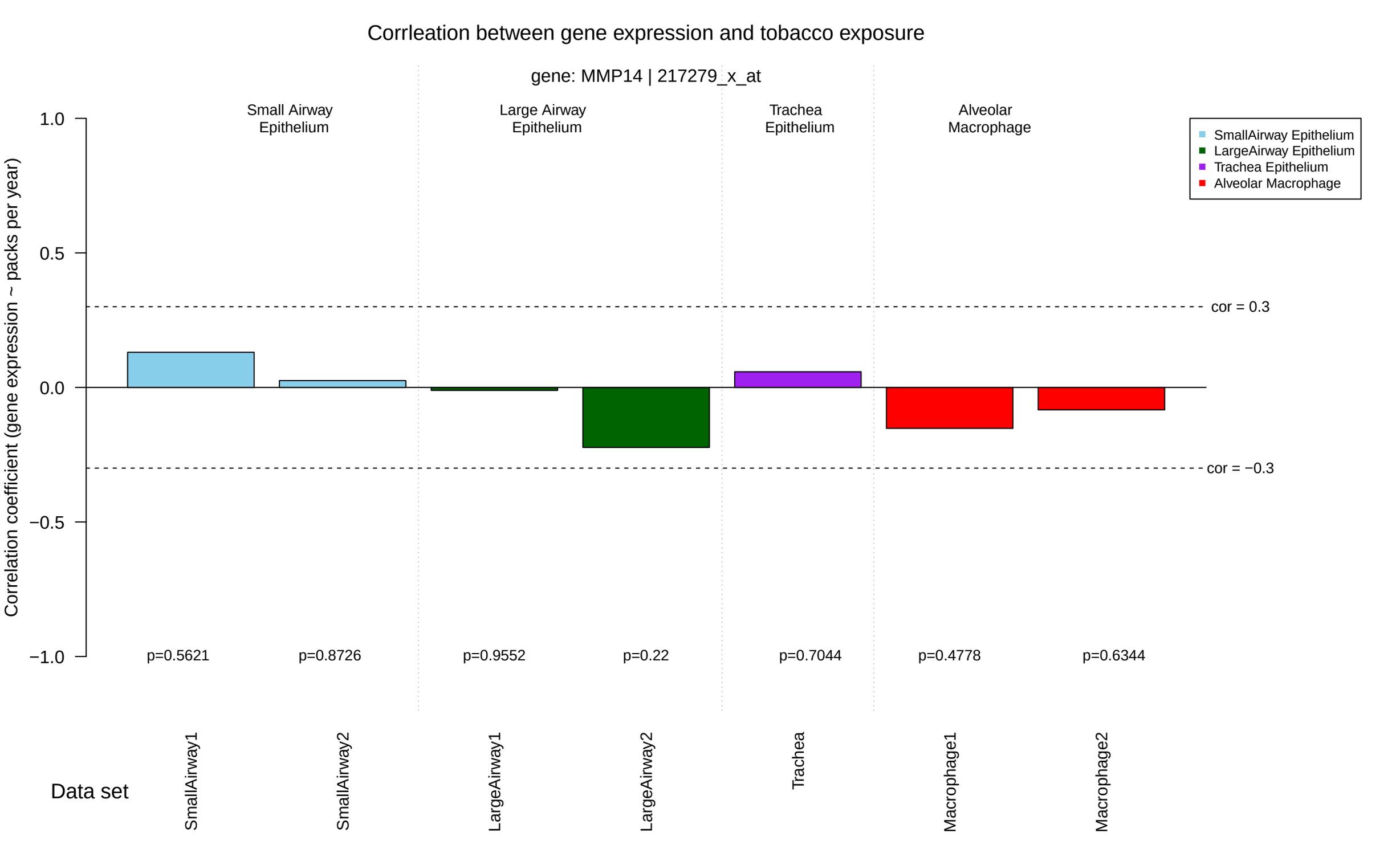

set: LargeAirway1 (p=0.409)
MMP14 | 217279_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.833)
MMP14 | 217279_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6192)
MMP14 | 217279_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0934)
MMP14 | 217279_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7371)
MMP14 | 217279_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MMP14 | 217279_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5621
p=0.8726
p=0.9552
p=0.22
p=0.7044
p=0.4778
p=0.6344
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2